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Exercise 1: Installing Your Monitoring Service

Student Information

Student Name: Afeez Lawal

Date: 25/05/25

Step 1 Initial Configuration:

Complete this section AFTER finishing step 1 of the assignment.

Nagios Installation and Configuration

Linux Distribution Used: UBUNTU

System IP Address:

Current IP Address: 10.0.2.15 (Before change)

Static IP Address Based on first name initial from Appendix A (e.g., A = .200, B = .201, etc.)

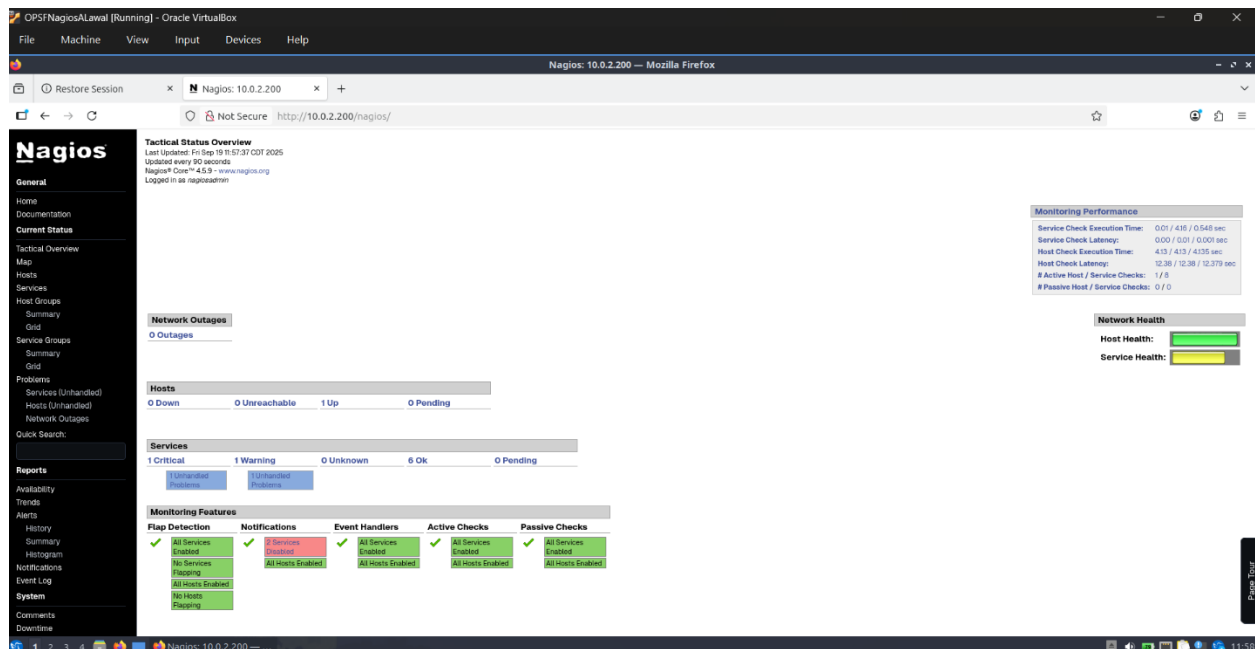
Static IP Address: 10.0.2.200 (After change)

Web Browser Access:

URL: <http://10.0.2.200/nagios/>

Login Credentials Setup: nagiosadmin and Bollyboy123# for Nagios access

Tactical Overview Screenshot:



Part 2: Script and Command Descriptions

Step 2: Command Interpretation

Script Commands:

List each command used during the installation and configuration of Nagios. See example below.

Command: `sudo apt-get update`

Description: Updates the package list for upgrades and new installations.

Continue this format for each command used, explaining the purpose and function of each.

Command: `sudo apt-get install -y autoconf gcc libc6 make wget unzip apache2 php libapache2-mod-php libgd-dev ufw`

Description: Installs the main tools and libraries Nagios needs (compiler, web server, PHP, graphics library, firewall utility).

Command: `sudo apt-get install -y openssl libssl-dev`

Description: This installs the extra tools and libraries Nagios plugins need (things like compilers, encryption libraries, SNMP support for monitoring devices, and math utilities).

Command: `cd /tmp`

Description: Switches to the temporary directory where we'll download the plugin files.

Command: `wget -O nagioscore.tar.gz $(wget -q -O - https://api.github.com/repos/NagiosEnterprises/nagioscore/releases/latest | grep '"browser_download_url":' | grep -o 'https://[^\"]*')`

Description: Downloads the latest version of the Nagios Plugins from GitHub into a compressed file.

Command: `tar xzf nagioscore.tar.gz`

Description: Extracts the compressed plugin files so they're ready to be installed.

Command: `cd /tmp/nagios-*`

Description: Moves into the nagios folder containing the plugin source code.

Command: `sudo ./configure --with-httpd-conf=/etc/apache2/sites-enabled`

Description: Prepares Nagios for your system and points it to Apache's config folder.

Command: `sudo make all`

Description: Compiles the Nagios program so it's ready to install..

Command: `sudo make install-groups-users`

Description: Creates a dedicated "nagios" user and group for security.

Command: `sudo usermod -a -G nagios www-data`

Description: Adds Apache's user account to the nagios group so the web interface can work.

Command: `sudo make install`

Description: Installs the Nagios program files onto the system.

Command: `sudo make install-daemoninit`

Description: Sets up Nagios as a service so it can run automatically at startup.

Command: `sudo make install-commandmode`

Description: Enables external commands so you can control Nagios from the web interface.

Command: `sudo make install-config`

Description: Installs example config files Nagios needs to start (can be customized later).

Command: `sudo make install-webconf`

Description: Installs Apache web settings for Nagios.

Command: `sudo a2enmod rewrite`

Description: Enables the rewrite module in Apache (needed for Nagios web pages).

Command: `sudo a2enmod cgi`

Description: Enables the CGI module in Apache (Nagios uses CGI scripts for its interface).

Command: `sudo ufw allow Apache`

Description: Allows web traffic (port 80) through the firewall.

Command: `sudo ufw reload`

Description: Reloads firewall rules so the changes take effect.

Command: `sudo htpasswd -c /usr/local/nagios/etc/htpasswd.users nagiosadmin`

Description: Creates the login account “nagiosadmin” with a password to access the web interface.

Command: `sudo systemctl restart apache2.service`

Description: Restarts Apache so it loads the new Nagios configuration.

Command: `sudo systemctl start nagios.service`

Description: Starts the Nagios monitoring service.

Command: `sudo apt-get install -y autoconf gcc libc6 libmcrypto-dev
make libssl-dev wget bc gawk dc build-essential snmp libnet-
snmp-perl gettext`

Description: Installs all the extra software packages and libraries Nagios plugins need to work (compilers, encryption support, SNMP tools, and math/text utilities)

Command: `wget -O nagios-plugins.tar.gz $(wget -q -O -
https://api.github.com/repos/nagios-plugins/nagios-
plugins/releases/latest | grep '"browser_download_url":' | grep
-o 'https://[^\"]*')`

Description: Downloads the latest Nagios Plugins package from GitHub into a compressed file.

Command: `tar xzf nagios-plugins.tar.gz`

Description: Extracts the compressed plugin package so the files are ready to install.

Command: `cd /tmp/nagios-plugins-*/`

Description: Moves into the folder where the plugin files were extracted.

Command: `sudo ./configure`

Description: Prepares the plugin installation for your system by checking dependencies and setting up paths.

Command: `sudo make`

Description: Compiles the plugin source code into executable programs Nagios can run.

Command: `sudo make install`

Description: Installs the compiled plugins onto the system so Nagios can start using them.

Exercise 2: Configure Monitoring to Enable Health Checks

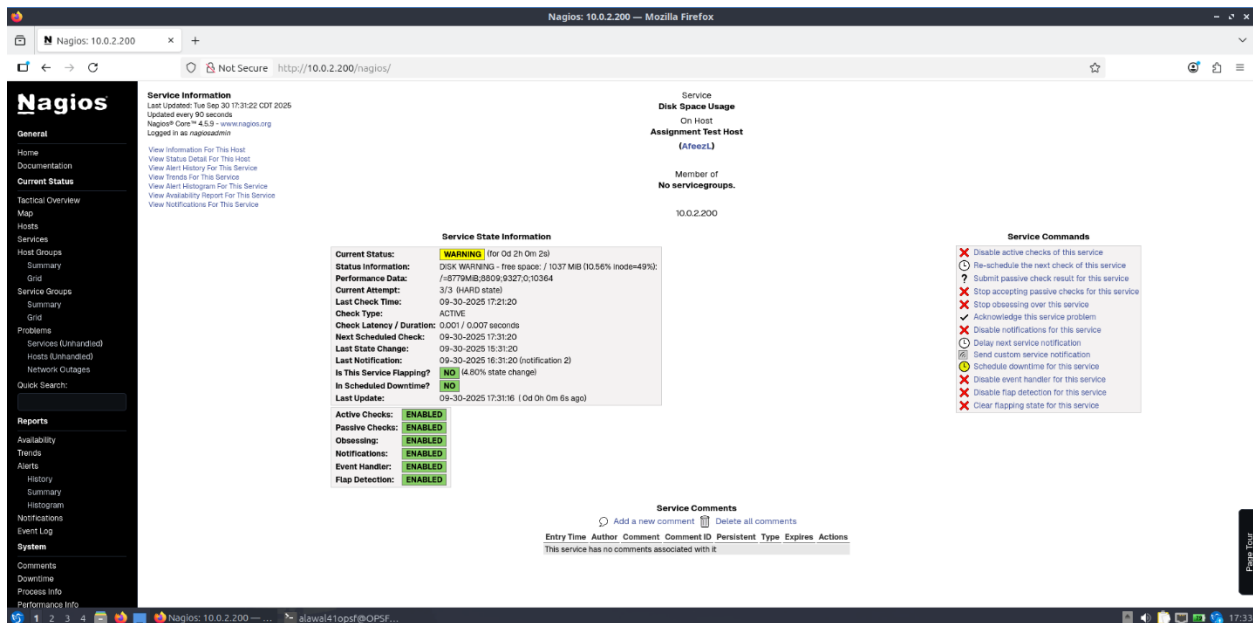
Student Information

Student Name: Afeez Lawal

Date: 1/10/25

Part 1: Screenshots of completed health checks

Figure 1: Disk Space Usage



Fi/usr/local/nagios/etc/objects/commands.cfg gure 2: Memory Usage

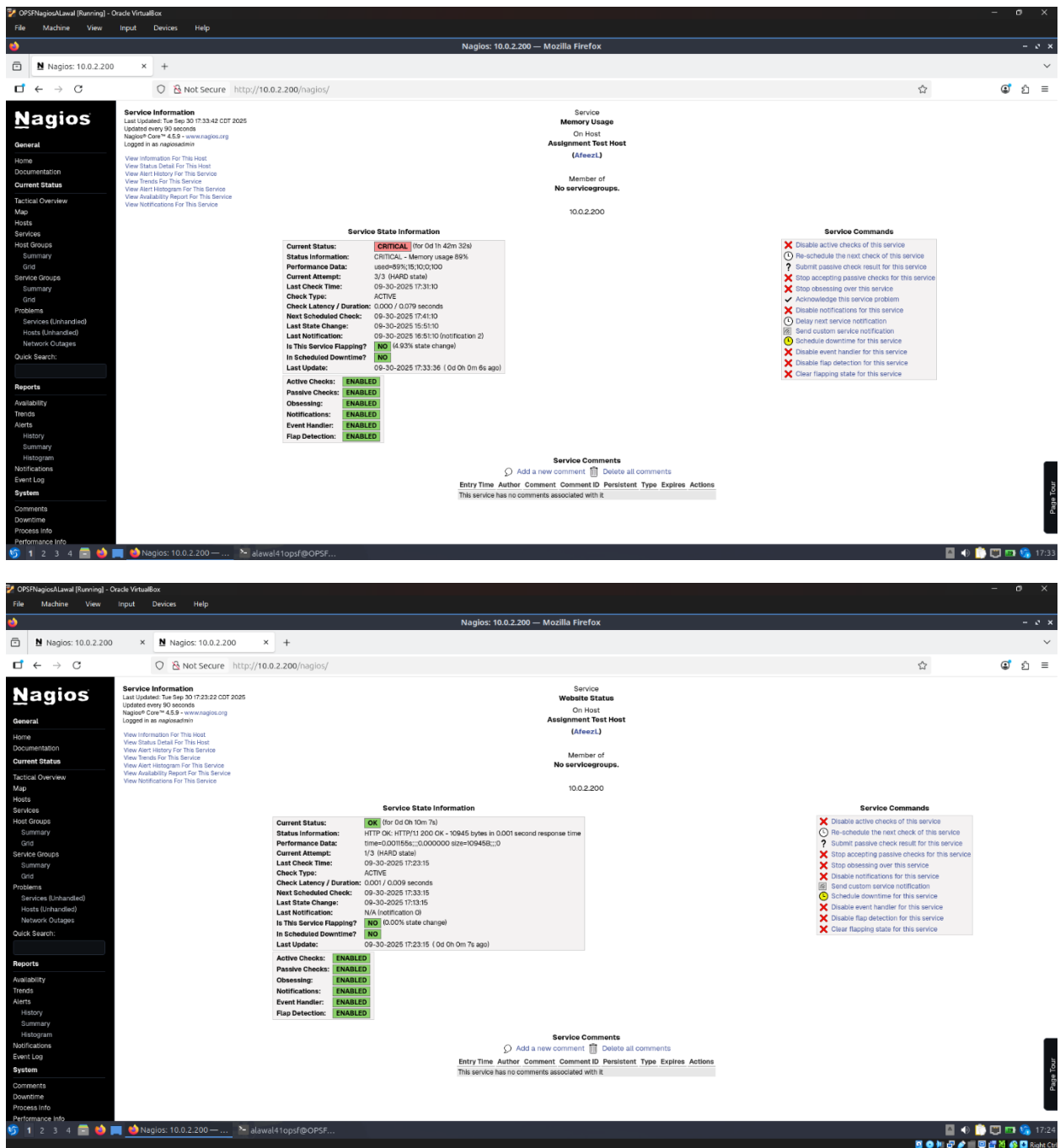


Figure 3: Website Status

Part 2: Configuration guide

Create your guide below, remember to site all of your sources and provide screenshots taken of your own work for each step that you are documenting.

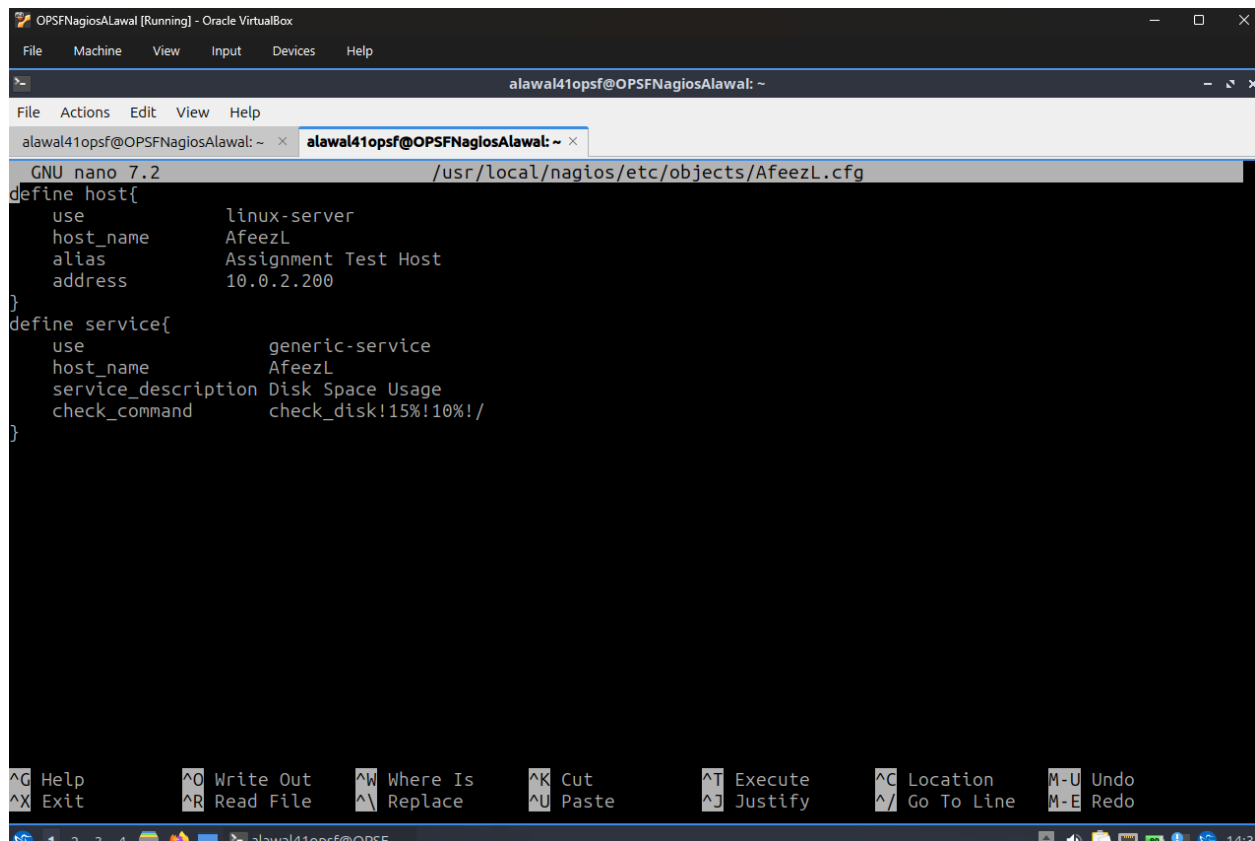
KB: Configuring Disk Space Monitoring in Nagios

Step 1: Add a service definition for disk usage

Inside my host configuration file(/usr/local/nagios/etc/objects/AfeezL.cfg), I added this block beneath the host definition:

- First I used: Sudo nano (/usr/local/nagios/etc/objects/AfeezL.cfg)

```
define service {  
    use          generic-service  
  
    host_name     AfeezL  
  
    service_description Disk Space Usage  
  
    check_command check_disk!15%!10%!/  
}
```



```
GNU nano 7.2 /usr/local/nagios/etc/objects/AfeezL.cfg
define host{
  use                linux-server
  host_name          AfeezL
  alias              Assignment Test Host
  address            10.0.2.200
}
define service{
  use                generic-service
  host_name          AfeezL
  service_description Disk Space Usage
  check_command       check_disk!15%!10%!/
}

^G Help      ^O Write Out  ^W Where Is   ^K Cut       ^T Execute   ^C Location  M-U Undo
^X Exit      ^R Read File  ^\ Replace    ^U Paste     ^J Justify   ^_ Go To Line M-E Redo
```

- service_description Disk Space Usage → label shown in Nagios web UI.
- check_command check_disk!15%!10%!/ → calls the check_disk plugin with thresholds:
 - Warning if less than 15% free.
 - Critical if less than 10% free.
 - / Specifies the root filesystem.

Step 2: Define the check_disk command

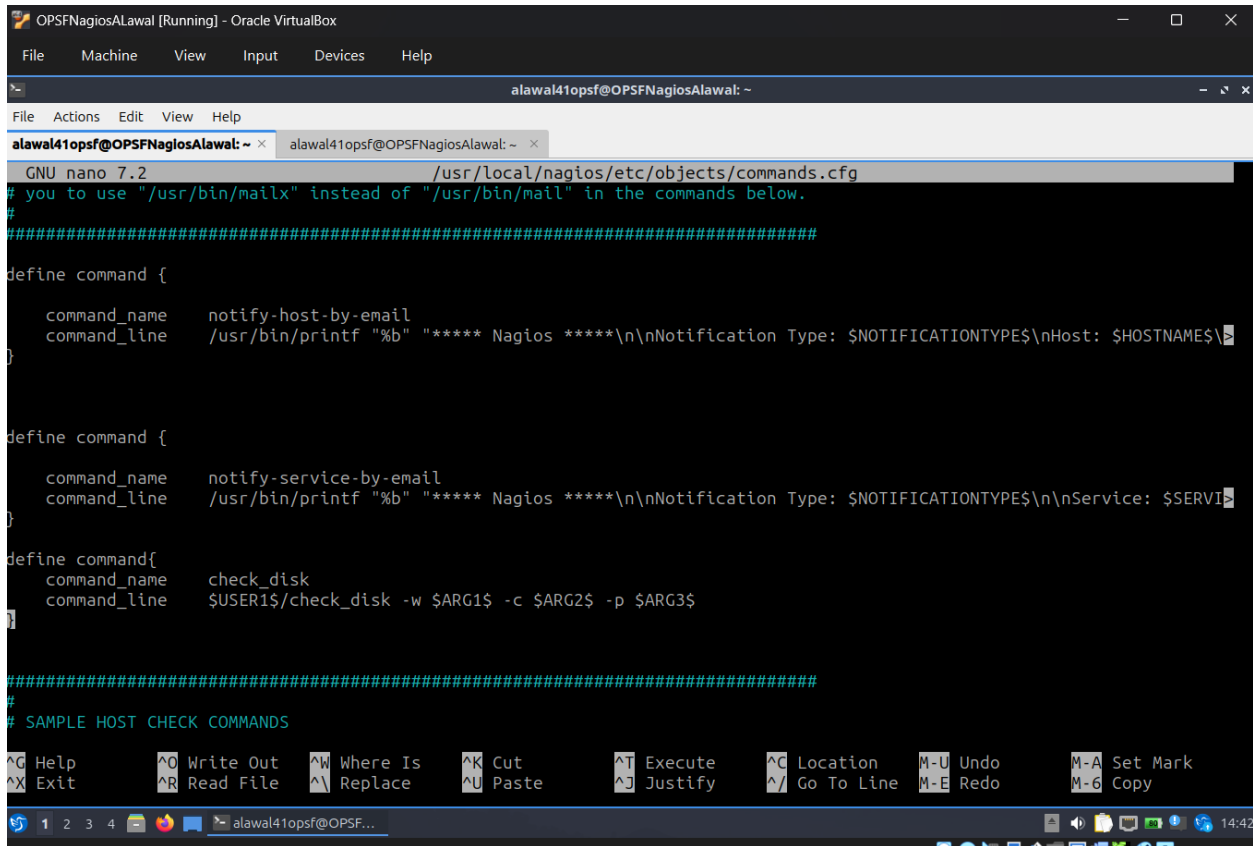
Nagios needs a command definition that maps the service call to the actual plugin. In /usr/local/nagios/etc/objects/commands.cfg I added:

- I used: Sudo nano (/usr/local/nagios/etc/objects/commands.cfg)

```
define command{
  command_name    check_disk
```

```
command_line $USER1$/check_disk -w $ARG1$ -c $ARG2$ -p $ARG3$  
}
```

This tells Nagios to run the check_disk plugin (stored in \$USER1\$, usually /usr/local/nagios/libexec) with the arguments passed in the service definition.



The screenshot shows a terminal window titled 'OPSFNagiosAlawal [Running] - Oracle VirtualBox'. The terminal is running the 'nano' text editor, editing the file '/usr/local/nagios/etc/objects/commands.cfg'. The file content is as follows:

```
GNU nano 7.2 /usr/local/nagios/etc/objects/commands.cfg  
# you to use "/usr/bin/mailx" instead of "/usr/bin/mail" in the commands below.  
#  
#####  
define command {  
    command_name    notify-host-by-email  
    command_line     /usr/bin/printf "%b" "***** Nagios *****\n\nNotification Type: $NOTIFICATIONTYPE$\nHost: $HOSTNAME$\nService: $SERVICE$\nDate: $DATE$\nTime: $TIME$\nAuthor: $AUTHOR$\nContact: $CONTACT$\nPriority: $PRIORITY$\nState: $STATE$\nOutput: $OUTPUT$\nExecuted: $EXECUTED$\nAcknowledged: $ACKNOWLEDGED$\nRepaired: $REPAIRED$\nResolved: $RESOLVED$\nComment: $COMMENT$\n" "$NOTIFICATIONTYPE$ $HOSTNAME$ $SERVICE$"  
}  
  
define command {  
    command_name    notify-service-by-email  
    command_line     /usr/bin/printf "%b" "***** Nagios *****\n\nNotification Type: $NOTIFICATIONTYPE$\nHost: $HOSTNAME$\nService: $SERVICE$\nDate: $DATE$\nTime: $TIME$\nAuthor: $AUTHOR$\nContact: $CONTACT$\nPriority: $PRIORITY$\nState: $STATE$\nOutput: $OUTPUT$\nExecuted: $EXECUTED$\nAcknowledged: $ACKNOWLEDGED$\nRepaired: $REPAIRED$\nResolved: $RESOLVED$\nComment: $COMMENT$\n" "$NOTIFICATIONTYPE$ $HOSTNAME$ $SERVICE$"  
}  
  
define command {  
    command_name    check_disk  
    command_line     $USER1$/check_disk -w $ARG1$ -c $ARG2$ -p $ARG3$  
}  
  
#####  
#  
# SAMPLE HOST CHECK COMMANDS
```

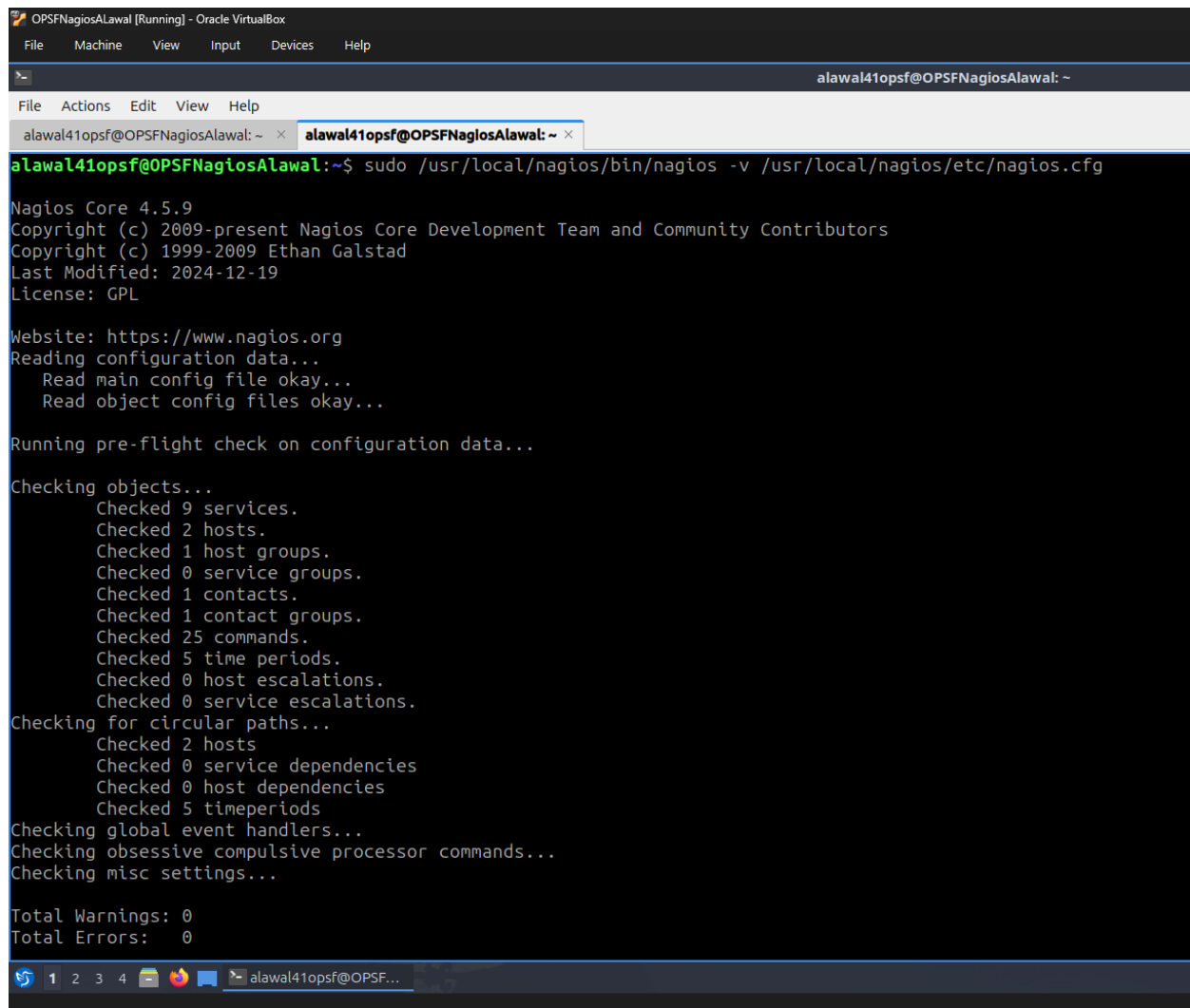
The terminal window also shows a menu bar with various editing and navigation options, and a status bar at the bottom indicating the current line and column.

Source: https://kifarunix.com/add-hosts-to-nagios-server-for-monitoring/?utm_source

Step 3: Verify the configuration

I ran the Nagios pre-flight check

(sudo /usr/local/nagios/bin/nagios -v /usr/local/nagios/etc/nagios.cfg)



```
alawal41opsf@OPSFNagiosAlawal: ~$ sudo /usr/local/nagios/bin/nagios -v /usr/local/nagios/etc/nagios.cfg

Nagios Core 4.5.9
Copyright (c) 2009-present Nagios Core Development Team and Community Contributors
Copyright (c) 1999-2009 Ethan Galstad
Last Modified: 2024-12-19
License: GPL

Website: https://www.nagios.org
Reading configuration data...
  Read main config file okay...
  Read object config files okay...

Running pre-flight check on configuration data...

Checking objects...
  Checked 9 services.
  Checked 2 hosts.
  Checked 1 host groups.
  Checked 0 service groups.
  Checked 1 contacts.
  Checked 1 contact groups.
  Checked 25 commands.
  Checked 5 time periods.
  Checked 0 host escalations.
  Checked 0 service escalations.
Checking for circular paths...
  Checked 2 hosts
  Checked 0 service dependencies
  Checked 0 host dependencies
  Checked 5 timeperiods
Checking global event handlers...
Checking obsessive compulsive processor commands...
Checking misc settings...

Total Warnings: 0
Total Errors: 0
```

The output confirmed there were **0 errors and 0 warnings**, meaning the configuration was valid. This is the standard validation process described in the <https://assets.nagios.com/downloads/nagioscore/docs/nagioscore/3/en/verifyconfig.html>

Step 4: Restart Nagios

Finally, I restarted Nagios to apply the new configuration: `sudo systemctl restart nagios/`
`sudo reboot.`

Sources:

- Justintime. (n.d.). *check_mem.pl – Nagios Memory Plugin*. GitHub. https://github.com/justintime/nagios-plugins/blob/master/check_mem/check_mem.pl
- Kifarunix. (2019, June 10). *Define Service Metric Checks for the Remote Hosts*. https://kifarunix.com/add-hosts-to-nagios-server-for-monitoring/?utm_source
- Nagios Plugins. (n.d.). *check_disk Plugin*. https://www.monitoring-plugins.org/doc/man/check_disk.html?utm_source
- Tutorialspoint. (n.d.). *Nagios – hosts and services*. https://www.tutorialspoint.com/nagios/nagios_hosts_and_services.htm?utm_source
- Packt Publishing. (2011). *Creating a new network host*. <https://subscription.packtpub.com/book/cloud-and-networking/9781849515566/1/ch01lvl1sec09/creating-a-new-network-host?utm>

Created/Modified By: Afeez Lawal

Date: 01/10/25

Notes/Changes:

Approved By:

Excerise 3: Knowledge Base - Configuring Memory Usage Monitoring in Nagios

Configure a Nagios health check to monitor Memory Usage on the host machine 'AfeezL', with thresholds set to Warning at 15% free memory and Critical at 10% free memory.

Step 1: Install the Memory Plugin

The default Nagios Plugins package does not include a memory check. We created a new plugin manually.

1. Create a new plugin file:

```
sudo nano /usr/local/nagios/libexec/check_mem.pl
```

2. Paste the check_mem.pl Perl script (reads /proc/meminfo to calculate memory %).

3. Save and exit, then make it executable:

```
sudo chmod +x /usr/local/nagios/libexec/check_mem.pl
```

4. Test manually to confirm it works:

```
/usr/local/nagios/libexec/check_mem.pl -w 15 -c 10
```

Example output:

```
CRITICAL - Memory usage 91% | used=91%;15;10;0;100
```

Step 2: Define the Command

Edit the Nagios commands.cfg file:

```
sudo nano /usr/local/nagios/etc/objects/commands.cfg
```

Add this definition:

```
define command{
    command_name check_mem
    command_line $USER1$/check_mem.pl -w $ARG1$ -c $ARG2$
}
```

Step 3: Add the Service Check

Edit the host configuration file AfeezL.cfg:

```
sudo nano /usr/local/nagios/etc/objects/AfeezL.cfg
```

Add the Memory Usage service:

```
define service{
    use generic-service
    host_name AfeezL
    service_description Memory Usage
    check_command check_mem!15!10
}
```

Step 4: Verify and Restart

Run a pre-flight check:

```
sudo /usr/local/nagios/bin/nagios -v /usr/local/nagios/etc/nagios.cfg
```

If there are no errors, restart Nagios:

```
sudo systemctl restart nagios
```

Step 5: Confirm in Web Interface

Open the Nagios web interface → Services → AfeezL.

Example output:

```
CRITICAL - Memory usage 91%
```

```
Performance Data: used=91%;15;10;0;100
```

This shows the plugin is working and reporting memory usage in percentages, as required.

References

Kifarunix. (2019, June 10). Define Service Metric Checks for the Remote Hosts.

<https://kifarunix.com/add-hosts-to-nagios-server-for-monitoring/>

Nagios Plugins. (n.d.). check_mem Plugin.

https://exchange.nagios.org/directory/Plugins/Operating-Systems/Linux/check_mem-2Epl/details

Tutorialspoint. (n.d.). Nagios – Hosts and Services.

https://www.tutorialspoint.com/nagios/nagios_hosts_and_services.htm

Exercise 4: Creating host in nagios

creating host in nagios

1) Find your nagios.cfg (so we know where to put the host file)

Run this and copy the result/path it prints:

- `sudo find / -name nagios.cfg 2>/dev/null | head -n 1`

2. Create the host file **sub-directory “AfeezL”** , to store your own objects –

- `sudo nano /usr/local/nagios/etc/objects/AfeezL.cfg`

2b. paste this code

-

```
define host{  
    use          linux-server  
  
    host_name    AfeezL  
  
    alias        Assignment Test Host  
  
    address      10.0.2.200  
  
}
```

https://www.tutorialspoint.com/nagios/nagios_hosts_and_services.htm?utm_source

https://kifarunix.com/add-hosts-to-nagios-server-for-monitoring/?utm_source

https://subscription.packtpub.com/book/cloud-and-networking/9781849515566/1/ch01lvl1sec09/creating-a-new-network-host?utm_source

- articles explain how to define hosts and what services are in Nagios

3. Telling Nagios server to load that host, Open main Nagios configuration file

- `sudo nano /usr/local/nagios/etc/nagios.cfg`

b. Scroll to the section where you see other `cfg_file` or `cfg_dir` lines. At the bottom of that section, and insert the line

- `cfg_file=/usr/local/nagios/etc/objects/AfeezL.cfg`

Save and exit.

4. Verify config

```
sudo /usr/local/nagios/bin/nagios -v /usr/local/nagios/etc/nagios.cfg
```

If it says **Total Warnings: 0 Total Errors: 0**, you're good.

KB: Adding a New Host in Nagios Core

Step 1: Locate the main Nagios configuration file

Before adding a host, it's important to know where Nagios loads its configuration from. We can search for the main config file with:

```
sudo find / -name nagios.cfg 2>/dev/null | head -n 1
```

This identifies the main nagios.cfg, which in my setup was /usr/local/nagios/etc/nagios.cfg.

- **Reference:** Kifarunix (2019) explains that host definitions must be placed in configuration files that are referenced by nagios.cfg, which acts as the entry point for all Nagios object definitions.
-

Step 2: Create a new host definition file

I created a new file called AfeezL.cfg in the objects directory:

```
sudo nano /usr/local/nagios/etc/objects/AfeezL.cfg
```

Inside, I added the following code:

```
define host{  
    use          linux-server  
  
    host_name     AfeezL  
  
    alias         Assignment Test Host  
  
    address       10.0.2.200  
}
```

- **Explanation:**
 - define host{...} begins a new host object definition.
 - use linux-server tells Nagios to inherit default settings from the linux-server template (defined in templates.cfg).
 - host_name is the unique identifier for the host in Nagios.
 - alias provides a human-friendly description.

- address is the IP address that Nagios will use to run checks (in this case, the VM IP 10.0.2.200).
 - **Reference:** Tutorialspoint (n.d.) shows an almost identical host block, with use linux-server, host_name, alias, and address. Packt (2011) further explains the meaning of each directive: host_name is the internal label, alias is a descriptive name, and address should ideally be an IP for reliability.
-

Step 3: Tell Nagios to load the host file

Next, I edited the main configuration file:

```
sudo nano /usr/local/nagios/etc/nagios.cfg
```

At the bottom of the section listing other cfg_file entries, I added:

```
cfg_file=/usr/local/nagios/etc/objects/AfeezL.cfg
```

This line tells Nagios to also load the definitions from the AfeezL.cfg file.

- **Reference:** The Nagios Support Knowledge Base (2015) article on *Adding a New Host and Service* shows that new object definition files must be included in nagios.cfg using the cfg_file directive.
-

Step 4: Verify the configuration

Finally, I verified the syntax to ensure there were no errors:

```
sudo /usr/local/nagios/bin/nagios -v /usr/local/nagios/etc/nagios.cfg
```

The output ended with:

Total Warnings: 0

Total Errors: 0

Things look okay - No serious problems were detected during the pre-flight check

This confirmed that the host was added correctly.

- **Reference:** Kifarunix (2019) also demonstrates using the nagios -v pre-flight check to validate changes before restarting the Nagios service.
- .

